Supplementary File 3. Screening Guide and Eligibility Decision Rules

Title

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1. Purpose

This screening guide provides operational rules for selecting evidence for the scoping review titled **“Reframing the malaria-sickle cell disease relationship in sub-Saharan Africa as a bounded syndemic: A scoping review and evidence gap map”** It is intended to support consistent screening decisions across reviewers during title and abstract screening and full-text review.

The guide translates the review’s Population, Concept, Context criteria into practical inclusion and exclusion rules. It also defines how to handle ambiguous studies, overlapping reports, grey literature, laboratory studies, conference abstracts, studies outside sub-Saharan Africa, and studies that address only part of the malaria-SCD syndemic framework.

2. Review focus

The review maps evidence on the relationship between malaria and sickle cell disease (SCD) in sub-Saharan Africa (SSA), using a bounded syndemic framework.

The review is concerned with seven evidence categories:

1. Co-occurrence
2. Spatial co-risk
3. Biological interaction
4. Clinical convergence

5. Service-mediated visibility

6. Structural vulnerability

7. Programme response

A record does not need to address all seven categories to be included. It must contribute meaningfully to at least one category and be relevant to the review question.

3. Main review question

How has the literature conceptualised, measured, and linked malaria-SCD biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, and screening-to-care systems in sub-Saharan Africa?

4. Screening principles

Reviewers should apply the following principles throughout screening.

1. Include broadly at title and abstract stage if relevance is plausible.

2. Exclude only when the record is clearly outside the review scope.

3. Use full-text screening to resolve uncertainty.

4. Do not exclude because the study does not use the word “syndemic.”

5. Do not include a record only because it mentions malaria and SCD unless it contributes evidence to at least one review category.

6. Treat co-occurrence as relevant but weak evidence.

7. Prioritise evidence that links biology, geography, clinical presentation, service visibility, structural access, or programme response.

8. Record all full-text exclusion reasons transparently.

5. Screening stages

5.1 Stage 1: Title and abstract screening

At title and abstract stage, each record should be coded as:

Decision	Meaning
Include	The record clearly meets the inclusion criteria or clearly contributes to one review category.
Exclude	The record clearly does not meet the inclusion criteria.
Unclear	The abstract does not provide enough information to decide.

Records coded as **Include** or **Unclear** by either reviewer should move to full-text screening.

5.2 Stage 2: Full-text screening

At full-text stage, reviewers should assess the complete source against the Population, Concept, Context criteria and the special decision rules. Each full text should be coded as:

Decision	Meaning
Include	The source meets the inclusion criteria and contributes to at least one evidence category.
Exclude	The source fails one or more eligibility criteria.
Discuss	The source is ambiguous and requires team adjudication.

For excluded full texts, reviewers must record one primary reason for exclusion.

6. Inclusion criteria

A record should be included if it meets all of the following criteria:

1. It addresses malaria, SCD, HbS, sickle cell trait, a malaria-relevant red blood cell polymorphism, severe anaemia, SCD under-recognition, newborn screening, SCD care, routine visibility, structural vulnerability, programme response, or syndemic theory.
2. It relates to SSA or malaria-endemic African health systems.
3. It has human, clinical, epidemiological, spatial, health-system, programme, policy, implementation, modelling, or theoretical relevance.
4. It provides sufficient information for charting.
5. It contributes evidence to at least one of the following categories: co-occurrence, spatial co-risk, biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, or programme response.

7. Exclusion criteria

A record should be excluded if it meets any of the following conditions:

1. It is outside SSA and has no direct conceptual, methodological, comparative, or intervention relevance to SSA.
2. It is a malaria study with no relevance to SCD, HbS, red blood cell polymorphisms, severe anaemia, clinical convergence, or syndemic interpretation.
3. It is an SCD study with no relevance to malaria-endemic settings, malaria exposure, under-recognition, newborn screening, service access, structural vulnerability, or programme planning.
4. It is an animal-only study with no direct relevance to human malaria-SCD relationships.
5. It is a laboratory-only study without clinical, epidemiological, population genetic, mechanistic, or programme relevance.

6. It focuses only on non-SCD haemoglobinopathies and does not inform malaria-relevant red blood cell variation.
7. It lacks sufficient information for data charting.
8. It is a duplicate report where a fuller or more relevant report is available.
9. It is an editorial, opinion piece, or commentary with no substantive conceptual, empirical, programme, or policy contribution.

8. Population screening rules

8.1 Include

Include studies involving:

1. Neonates, infants, children, adolescents, adults, or pregnant women in SSA where relevant to SCD burden, malaria exposure, or child health systems.
2. Individuals with SCD, SCA, HbSS, HbAS, sickle cell trait, or HbS.
3. Populations carrying malaria-relevant red blood cell polymorphisms, including HbC, alpha-thalassaemia, G6PD deficiency, ABO blood group, Dantu, or related erythrocyte variants.
4. Caregivers of children with SCD.
5. Health workers, laboratories, facilities, newborn screening programmes, transfusion services, routine health information systems, or referral networks relevant to SCD or malaria-SCD care.

8.2 Exclude

Exclude studies involving:

1. Animal-only populations with no direct human relevance.

2. Populations outside SSA with no direct relevance to the review.

3. Human populations where neither malaria, SCD, HbS, red blood cell polymorphisms, severe anaemia, screening, care access, nor syndemic theory is relevant.

9. Concept screening rules

9.1 Biological interaction

Include if the study addresses:

1. HbAS protection against malaria.

2. HbS persistence or malaria selection.

3. HbSS or SCA burden linked to malaria-endemic populations.

4. HbC, alpha-thalassaemia, G6PD deficiency, ABO blood group, Dantu, or other red blood cell polymorphisms in relation to malaria.

5. Epistasis among malaria-relevant red blood cell polymorphisms.

6. Mechanisms such as parasite invasion, rosetting, cytoadherence, red cell membrane changes, parasite growth, or malaria phenotype modification.

Exclude if the study:

1. Discusses red blood cells without malaria or SCD relevance.

2. Is laboratory-only and does not inform human malaria-SCD interpretation.

3. Addresses non-malaria genetic traits with no relevance to the review framework.

9.2 Spatial co-risk

Include if the study addresses:

1. Malaria exposure, prevalence, incidence, mortality, or endemicity maps.

2. HbS distribution, SCD prevalence, or SCA birth-burden mapping.

3. Spatial heterogeneity of malaria, HbS, or SCD.

4. Co-risk, co-clustering, spatial overlap, geographic accessibility, facility catchments, or spatial prioritisation.

5. Subnational variation in sickle-related risk or malaria burden.

Exclude if the study:

1. Uses geography only as a location label without spatial analysis or planning relevance.

2. Maps a disease unrelated to malaria, SCD, HbS, severe anaemia, or service access.

9.3 Clinical convergence

Include if the study addresses:

1. Severe anaemia.

2. Malaria-positive illness in children with SCD or SCA.

3. Severe malaria phenotypes among children with HbS, HbAS, HbSS, or SCA.

4. Acute febrile illness, respiratory distress, transfusion need, or paediatric deterioration where malaria and SCD may overlap.

5. Diagnostic attribution where malaria, severe anaemia, and SCD may be confused or co-recorded.

6. Hidden SCA among children admitted with severe anaemia or malaria-like illness.

Exclude if the study:

1. Describes malaria clinical features with no relevance to SCD, HbS, severe anaemia, or red blood cell polymorphisms.

2. Describes SCD clinical care with no relevance to malaria-endemic settings, severe anaemia, under-recognition, or service planning.

9.4 Service-mediated visibility

Include if the study addresses:

1. Newborn screening.

2. Confirmatory diagnosis.

3. Clinical recognition or missed diagnosis.

4. Routine data, DHIS2, HMIS, registers, surveillance, coding, or reporting.

5. Hospital admission data.

6. Mortality attribution or cause of death.

7. Service-visible burden or hidden burden.

8. Referral, clinic enrolment, retention, or follow-up as mechanisms of visibility.

Exclude if the study:

1. Uses routine data without relevance to malaria, SCD, diagnosis, reporting, or service visibility.

2. Reports administrative data with no disease or programme relevance.

9.5 Structural vulnerability

Include if the study addresses:

1. Unequal access to newborn screening, confirmatory diagnosis, counselling, prophylaxis, vaccination, hydroxyurea, transfusion, specialist care, referral, or follow-up.

2. Geography, transport, cost, workforce, laboratory access, medicine availability, or blood supply as barriers to SCD care.

3. Structural drivers of under-recognition, mortality, or poor outcomes.

4. Facility readiness or service availability for SCD care.

Exclude if the study:

1. Discusses social determinants generally without malaria, SCD, severe anaemia, screening, or care relevance.

2. Discusses health systems without connection to the review concepts.

9.6 Programme response

Include if the study addresses:

1. Newborn screening programmes.

2. Point-of-care testing.

3. Screening-to-care linkage.

4. Caregiver notification and counselling.

5. Clinic enrolment and follow-up.

6. Early standardised care.

7. Decentralised care, referral networks, hub-and-spoke models, or implementation networks.

8. Programme scale-up, sustainability, or service strengthening for SCD in SSA.

Exclude if the programme does not address SCD, malaria, screening, care linkage, or service response.

9.7 Syndemic theory

Include if the study:

1. Explicitly uses syndemic theory.

2. Uses syndemic terminology.

3. Examines disease clustering, biological interaction, structural amplification, or excess burden in ways relevant to malaria-SCD.

4. Does not use the word syndemic but links malaria, SCD, clinical burden, structural vulnerability, or programme response in a syndemic-relevant way.

Exclude if the study uses syndemic terminology but has no relevance to malaria, SCD, SSA, or the review framework.

10. Context screening rules

10.1 Include

Include studies from:

1. SSA countries.

2. Multicountry African studies that include SSA.

3. Malaria-endemic African health systems.

4. Community, hospital, laboratory, programme, policy, district, regional, national, or multicountry settings relevant to the review.

10.2 Exclude

Exclude studies from:

1. Non-SSA settings with no direct relevance to the review.

2. Global studies where SSA results cannot be identified and the study does not provide relevant conceptual or methodological evidence.

3. Settings unrelated to malaria-endemic African health systems.

11. Special decision rules

11.1 Studies outside SSA

Include outside-SSA studies only when they provide foundational or directly transferable evidence relevant to the review.

Examples that may be included:

1. Foundational penicillin prophylaxis evidence relevant to essential SCD care.

2. Syndemic theory or methods papers.
3. Foundational mechanistic studies explaining malaria-red cell interaction.
4. Methodological papers relevant to scoping reviews or syndemic classification.

Exclude outside-SSA studies when:

1. They describe SCD care in high-income settings with no transferability to SSA.
2. They do not address malaria, SCD, red cell polymorphisms, screening, service access, or syndemic theory.

11.2 Laboratory studies

Include laboratory studies if they explain mechanisms relevant to the review, including:

1. HbAS protection.
2. Parasite invasion.
3. Rosetting.
4. Cytoadherence.
5. Erythrocyte membrane biology.
6. Red cell polymorphism effects.
7. Epistasis.

Exclude laboratory studies if they have no clear human, epidemiological, clinical, or programme relevance.

11.3 Malaria-only studies

Include malaria-only studies if they address:

1. HbS or sickle cell trait.

2. Red blood cell polymorphisms.

3. Severe anaemia.

4. Malaria biomarkers relevant to SCD interpretation.

5. Malaria mapping relevant to co-risk.

6. Malaria data systems relevant to routine visibility.

Exclude malaria-only studies if they do not connect to any review concept.

11.4 SCD-only studies

Include SCD-only studies if they address:

1. SSA or malaria-endemic African settings.

2. SCD prevalence or burden.

3. Newborn screening.

4. Under-recognition.

5. Severe anaemia.

6. Hydroxyurea, transfusion, prophylaxis, vaccination, referral, or follow-up.

7. Routine data, coding, reporting, or mortality attribution.

8. Structural vulnerability or service access.

Exclude SCD-only studies if they have no relevance to SSA, malaria-endemic systems, service access, programme response, or the syndemic framework.

11.5 Studies on other haemoglobinopathies or red cell disorders

Include if the study addresses malaria-relevant red blood cell variation, such as HbC, alpha-thalassaemia, G6PD deficiency, ABO blood group, Dantu, or epistasis.

Exclude if the study focuses on unrelated haematological conditions without malaria or SCD relevance.

11.6 Reviews

Include reviews if they synthesize evidence relevant to the review question or one of the domains.

Screen primary studies cited in included reviews if they appear directly relevant.

11.7 Commentaries and Opinion Pieces

Include only if they provide substantive conceptual, theoretical, policy, or programme insight.

Exclude commentaries that simply state an opinion without evidence, framework, or policy relevance.

11.8 Conference abstracts

Include only if the abstract provides sufficient detail on:

1. Population.

2. Setting.

3. Methods.

4. Main finding.

5. Relevance to at least one review domain.

Exclude abstracts with insufficient information for charting.

11.9 Duplicate or overlapping reports

If two reports use the same data:

1. Retain the most complete source for general extraction.

2. Retain secondary reports if they provide distinct domain-specific evidence.

3. Document overlap in the data charting form.

12. Full-text exclusion reasons

At full-text stage, reviewers should assign one primary exclusion reason.

Exclusion reason	**Definition**
Wrong geography	Not SSA and no relevant conceptual, methodological, intervention, or comparative contribution.
Wrong population	Population does not relate to malaria, SCD, HbS, red cell polymorphisms, severe anaemia, screening, or care systems.
Wrong concept	Does not address any review concept.
No malaria relevance	Does not address malaria, malaria-endemic context, malaria biology, malaria burden, or malaria-SCD interpretation.

No SCD or HbS relevance	Does not address SCD, SCA, HbS, HbAS, sickle cell trait, or SCD care.
No syndemic relevance	Does not contribute to any evidence category.
Laboratory-only without review relevance	Laboratory study with no human, clinical, epidemiological, mechanistic, or programme relevance.
Insufficient information	Not enough detail to chart source.
Duplicate report	Same evidence reported more fully elsewhere.
Commentary without substantive contribution	Opinion source without conceptual, empirical, policy, or programme value.

14. Reviewer calibration procedure

14.1 Calibration before Title and Abstract screening

Before formal screening begins:

1. All reviewers will receive the protocol and screening guide.
2. A pilot sample of records will be screened independently.
3. Reviewers will compare decisions.
4. Disagreements will be discussed.
5. Eligibility rules will be clarified if needed.
6. The screening guide will be updated if major ambiguity is identified.

14.2 Calibration before Full-Text screening

Before full-text screening:

1. A sample of full texts will be screened independently.

2. Reviewers will compare inclusion and exclusion decisions.

3. Disagreements will be discussed.

4. Full-text exclusion categories will be refined if needed.

14.3 Calibration before Data charting

Before full data charting:

1. A small sample of included studies will be charted independently.

2. Reviewers will compare extracted fields.

3. Differences in interpretation will be resolved.

4. Coding definitions will be clarified.

5. The data charting form and codebook will be updated if needed.

15. Screening conflict resolution

Disagreements will be resolved using the following sequence:

1. Reviewers discuss the record and compare reasons.

2. Reviewers return to the eligibility criteria and decision rules.

3. If disagreement remains, the record is referred to the review lead.

4. If needed, a third reviewer or domain expert adjudicates.

5. The final decision and reason are documented.

16. Screening decision form

16.1 Title and Abstract screening form

Field	Response options
Study ID	Free text
Reviewer	Free text
Title	Auto-filled or free text
Abstract available	Yes, No
Potential SSA relevance	Yes, No, Unclear
Potential malaria relevance	Yes, No, Unclear
Potential SCD, HbS, or red cell relevance	Yes, No, Unclear
Potential clinical, service, structural, programme, or theory relevance	Yes, No, Unclear
Screening decision	Include, Exclude, Unclear
Reviewer notes	Free text

16.2 Full-text screening form

Field	Response options
Study ID	Free text
Reviewer	Free text
Full text available	Yes, No
Population eligible	Yes, No, Unclear
Concept eligible	Yes, No, Unclear
Context eligible	Yes, No, Unclear
Evidence source type eligible	Yes, No, Unclear
Contributes to at least one domain	Yes, No, Unclear

Domains identified	Co-occurrence, spatial co-risk, biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, programme response, syndemic theory
Final decision	Include, Exclude, Discuss
Primary exclusion reason	Select from full-text exclusion reasons
Reviewer notes	Free text

17. Minimum inclusion threshold

A study must meet the minimum inclusion threshold below:

1. It is relevant to SSA or malaria-endemic African health systems.
2. It addresses malaria, SCD, HbS, red cell polymorphisms, severe anaemia, routine visibility, service access, programme response, or syndemic theory.
3. It contributes to at least one review domain.
4. It has enough detail for charting.

If any of these are absent, exclude at full-text stage.

18. Domain coding during Full-text screening

During full-text screening, reviewers should assign preliminary domain codes. These codes may be refined during data charting.

Domain code	**Use when the study addresses**
D1 Biological interaction	HbAS protection, HbS persistence, red cell polymorphisms, malaria mechanisms, epistasis.

D2 Spatial co-risk	Malaria maps, HbS maps, SCA birth burden, spatial overlap, co-clustering, geographic access.
D3 Clinical convergence	Severe anaemia, malaria-positive illness, transfusion need, respiratory distress, under-recognised SCA in clinical settings.
D4 Service-mediated visibility	Screening, diagnosis, routine reporting, coding, mortality attribution, hidden burden, service-visible burden.
D5 Structural vulnerability	Access to diagnosis, prophylaxis, hydroxyurea, transfusion, specialist care, referral, retention.
D6 Programme response	Newborn screening, point-of-care testing, screening-to-care linkage, decentralised care, implementation networks.
D7 Syndemic theory	Explicit syndemic theory, implicit syndemic logic, structural amplification, co-occurrence versus interaction.

19. Notes for reviewers

1. Do not exclude a study because it uses “anaemia” rather than “sickle cell disease” if the abstract suggests possible SCD, HbS, severe malaria, or severe anaemia relevance. Move to full text if uncertain.

2. Do not exclude a study because it does not mention “syndemic.” Most relevant studies may not use syndemic terminology.

3. Do not include a study only because it mentions Africa. It must contribute to one review domain.

4. Do not include a study only because it mentions malaria. It must connect to SCD, HbS, red cell polymorphisms, severe anaemia, service systems, or the review framework.

5. Do not include a study only because it mentions SCD. It must relate to SSA, malaria-endemic systems, under-recognition, screening, care access, structural vulnerability, or programme response.
6. When unsure, code "Unclear" at title and abstract stage and move to full text.
7. At full-text stage, include only when the study clearly meets the minimum inclusion threshold.

20. Output of screening

Screening will produce:

1. A deduplicated record set.
2. Title and abstract screening decisions.
3. Full-text screening decisions.
4. A list of included studies.
5. A table of excluded full texts with reasons.
6. Preliminary domain codes for included studies.
7. A PRISMA-ScR flow diagram.

21. Version control

Any changes to this screening guide will be recorded in a version-control log.

Version	Date	Change made	Reason	Approved by
1.0	To be inserted	Initial screening guide	Initial version	To be inserted

1.1	To be inserted	To be inserted	To be inserted	To be inserted